

GUARDIAN

OT Network Threat Detection Platform Security Assessment Report

Report Date	June 15, 2026 15:50
	OT/ICS Security Assessment
Protocols Tested	Modbus TCP, OPC-UA, DNP3
	0
Critical Alerts	0
	v1.0

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1. Executive Summary

The GUARDIAN OT Threat Detection Platform conducted a comprehensive security assessment of the industrial control system network. During the assessment period, GUARDIAN detected **0 security events** across multiple OT protocols including Modbus TCP and OPC-UA.

The assessment revealed significant security gaps in the OT network, including unauthorized write commands to PLC registers, rogue device detection, replay attacks, and out-of-range value injection. A total of **0 CRITICAL** severity alerts were generated, indicating immediate risk to plant operations and safety.



Key Findings:

■	CRITICAL	Modbus TCP has no authentication — any device can write to PLC registers
■	CRITICAL	Safety system registers accessible without authorization
■	HIGH	Replay attacks possible — no sequence number validation in Modbus
■	HIGH	Rogue devices can join OT network without detection
■	MEDIUM	Excessive polling detected — attacker mapping process parameters
■	LOW	OPC-UA anonymous access enabled — process data exposed

2. Alert Statistics

2.1 Alerts by Severity

Severity	Count	Percentage	Risk Level
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2.2 Detection Rules Triggered

Rule	Count	Severity	MITRE TTP
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2.3 Top Attacking IPs

Rank	IP Address	Alert Count	Threat Level
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3. Attack Timeline

Chronological record of all detected security events:

Timestamp	Severity	Rule	Source IP	MITRE
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4. OT Asset Impact Analysis

GUARDIAN's Asset Impact Analyzer maps every attack to physical OT assets and real-world consequences:

Asset	Attack Type	Safety Risk	Risk Score	Recovery Time
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5. MITRE ATT&CK; for ICS Mapping

All detected attacks mapped to MITRE ATT&CK; for ICS framework techniques:

Detection Rule	Tactic	Technique ID	Technique Name	Severity
UNAUTHORIZED_WRITE	Impair Process Control	T0855	Unauthorized Command Message	CRITICAL
ROGUE_DEVICE	Initial Access	T0846	Remote System Discovery	HIGH
REPLAY_ATTACK	Evasion	T0856	Spoof Reporting Message	HIGH
EXCESSIVE_POLLING	Discovery	T0840	Network Connection Enumeration	MEDIUM
VALUE_INJECTION	Impair Process Control	T0855	Unauthorized Command Message	CRITICAL
HONEYPOT_TRIGGERED	Collection	T0843	Program Download	CRITICAL

6. Remediation Roadmap

6.1 Immediate Actions (0-7 days)

#	Action	Priority	Effort
1	Implement Modbus TCP authentication using vendor-specific security modules	CRITICAL	Medium
2	Enable OPC-UA security mode — disable anonymous access	CRITICAL	Low
3	Deploy network segmentation — isolate OT from IT network	CRITICAL	High
4	Implement IP whitelisting on all Modbus devices	HIGH	Low
5	Enable audit logging on all PLC and SCADA systems	HIGH	Low

6.2 Short Term (7-30 days)

#	Action	Priority	Effort
1	Deploy IDS/IPS solution for OT network monitoring	HIGH	High
2	Implement Modbus sequence number validation	HIGH	Medium
3	Deploy honeypot registers for early attacker detection	HIGH	Low
4	Conduct OT security awareness training for operators	MEDIUM	Medium
5	Implement automated backup for all PLC programs	MEDIUM	Low

6.3 Long Term (30-90 days)

#	Action	Priority	Effort
1	Implement IEC 62443 compliance framework across OT network	HIGH	Very High
2	Deploy encrypted communications for all OT protocols	HIGH	High
3	Implement Security Information and Event Management (SIEM)	MEDIUM	High
4	Conduct annual OT penetration testing exercise	MEDIUM	High
5	Develop and test OT incident response playbook	MEDIUM	Medium